Alex — Tier 1 sequential model outline

Role of this document

This note is the **advisor-ordered spine**: each section follows the sequence Alex emphasized (minimal model → assumptions → data → fitting → what we extract). It **does not replace** existing work; it points to it.

Paper Directions 11 (pool pre-sorting & promotion rules) is **§6A** — separate from the **538** empirical ladder (§4–§6). Read §6A before treating **538** as addressing "how teams are built." **Printable design note:** `sports/documents/Tier1_Presorting_Design_Note.md`.

Stable references (do not fork their content here):

- `Tier1_Briefing_Outline.md` — full technical detail, equations, and data column map.
- `Tier1_Narrative_Outline.md` — narrative arc and voice.
- `sports/documents/tier_1_roadmap.md` — execution contract for `535` / panel columns.
- **`sports/538_alex_tier1_model_and_fit.ipynb`** — notebook that should **mirror this section order** for code + outputs (`537` stays the simulation lab). **Cell-level map: §10 below** (update when 538 changes).
- `sports/documents/tier_1_roadmap.md` — **535 execution contract** (pipeline steps, `df` columns, CELL 0 switches for `535_sports_tier_1.ipynb`). Not superseded by 538; 538 reuses the same pipeline modules with a slimmer cell ladder.
- **Wang-style cross-domain template** (same *logical* steps as this outline): `Vector_Questions_and_Modeling_Thoughts.md` §11; mechanics of the Yin–Wang failure/success paper: `wang_paper_model.md`.

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1. Unit of modeling

- State the primary unit (i, j, t) and the empirical row (player–season / player–team–season).
- One sentence: local pool vs global advancement.

Fill from `Tier1_Briefing_Outline.md` §1; keep only what you will say aloud.

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2. Minimal model objects (headline only)

- Outcome Y (global advancement).
- Amalgamated local environment L_{ijt} (Tier 1 object); first proxy Q.
- Global selection capacity Λ_t.
- Role of own performance / ability proxy A_{ijt} (baseline, not the mechanism headline).

Detail table: `Tier1_Briefing_Outline.md` §2–§3.

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3. Minimal assumptions

- Local vs global scarcity framing.
- Leave-self-out for peer-based Q.
- Variable domains (binary Y, standardized L for fitting, etc.).
- What Tier 1 **defers** (sorting, dynamics, full causal ID). **How pools are *formed*** is the PD11 generative agenda — see **§6A**; §4–§6 **take pools as given** in the data.

Bullets only here; expand in manuscript later.

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4. Map model to data

One table: object → meaning → `df` column / construction step (`530` / panel load + Tier 1 cols).

Canonical table: `Tier1_Briefing_Outline.md` §4.

In `538`: CELL 1 (`CFG`, panel path) → **CELL 2** (`load_panel`, `apply_perf_metric_for_analysis`, legacy `poolq_loo`) → **CELL 3** (`add_tier1_mechanism_variables`, `ALEX_L_COL` from `PRIMARY_POOL_MODE`).

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5. Primary estimand

- **Main object to report:** turning point L^* (or Q^* in raw units), not only linear coefficients.
- One sentence on why (competing margins / inverted-U readout).

Formulas: `Tier1_Briefing_Outline.md` §7.

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6. Fitting plan (ordered)

1. Descriptive bins (no model). → `538` **CELL 4** (table), **CELL 4A** (binned line; optional if CELL 5 overlay is enough).
2. Transparent quadratic spec in L (LPM) + controls as needed. → `538` **CELL 5** (OLS on z-scored L, L^2, A; binned means + fitted curve).
3. Binary response (logit/probit) for the same index structure. → `538` **CELL 6** (logit; same index; L^* from linear part).
4. Robustness: FE, clustering, alternative L proxy, one decomposition term at a time. → `538` **CELL 7+** (placeholder).

Exploratory (not Alex §6 step 1; from Paper Directions 11 meeting): mean peer Q × LOO peer perf **SD** vs draft. → `538` **CELL 4B** (hexbin + Q histogram), **CELL 4C** (3D surface; optional). Always uses `congestion_quality` on *x* even when `PRIMARY_POOL_MODE='crowding'`.

Expand: `Tier1_Briefing_Outline.md` §5–§6.

6A. Paper Directions 11 — pool formation, promotion rules, and where they live

Source: `transcripts/20260515_Paper_Directions_11_otter_ai_transcript.pdf` (2026-05-15).
Do not confuse with §6: §6 is the **Wang-style empirical ladder** on **realized** pools. PD11 is about **questioning the rules** that put people in pools and that turn pool context into promotion — especially **pre-sorting / team building** in a **generative** layer (planned `538`; `537` **frozen**), not re-running more parameter grids.

What PD11 is *not* asking for (here)

- Replacing §6 with new bins/LPM/logit specs alone.
- Calibrating the generative model by **copying observed college team means** — Alex: misleading for a **multi-domain** minimal story (forensic checks on overlap/inequality across domains are fine).
- Treating **538** as if it had already solved pool formation — it has not (see **Where each thread lives** below).

Three threads (meeting order of emphasis)

#	Thread	Question	Priority in code
A	**Pre-sorting / how teams (pools) are built**	How do we **assign** individuals to local pools so pools **overlap** in talent, are not rigidly ordered slices, and **# pools ≫ # quantile bins**?	`538` (planned generative cells + `tier1_sim_config.py`). `537` **frozen** (legacy sort-and-chop in `assign_pool_ids`). Forensics: `530` **CELL 8–9.**
B	**Pool mean × pool dispersion**	Does advancement depend on LOO **mean** peer ability *and* LOO **SD** of peer ability (crowding vs distinction)?	`538` **CELL 4B, 4C** (exploratory EDA). Columns: `congestion_quality`, `peer_perf_sd_loo` (`tier1_mechanism_vars.py`).
C	**Promotion score vs local rank**	Should promotion weight **gap to pool mean** ($A_i - \bar{A}_{\text{pool}}$) rather than **rank within pool** (same rank whether peers are tight or spread)?	`537` first — e.g. `ADDITIVE_LOCAL_RANK_WEIGHT` / score mode in `sim_config.py`, Cell 10 playground. **Then** revisit **A** if shape does not move.

Alex's implementation order (transcript): tweak **C** in existing simulation machinery first (easy); if insufficient, rebuild **A**; use **B** in data to see whether dispersion belongs in the story.

Thread A — pre-sorting (generative pool assignment)

Problem with the old rule (still in `537` for assortative pools):

1. Draw abilities for all i.
2. Sort (optionally add Gaussian ε to a **sorting signal**).
3. **Chop** the sorted list into J equal-sized pools → **hard, ordered** teams with little realistic overlap.

Critique (PD11): teams look like **non-overlapping talent intervals**; ε-smearing is **too linear** for **heavy tails** (elite player on weak team occasionally); **# teams must be much larger than # quantile bins** used in EDA.

Directions discussed (generative — pick one family to prototype in `538`):

1. **Target means, then soft assignment (preferred framing)**
   - Draw a **target mean ability T_j** for each pool j (e.g. J draws from Uniform$[0, 1]$ or another spread).
   - Assign i with ability α_i to pool j with probability **↑ as α_i is close to T_j** (Gaussian kernel around T_j, or heavier tail).
   - Yields **overlapping "talent windows"** across pools — some tight, some spread out.
2. **Sequential / preferential assignment**
   - Start with even assignment probabilities; **update** as pools fill (Barabási-style "rich get richer") — another way to avoid rigid slices.
3. **Heavy-tailed reassignment noise**
   - Charles: ε from a **heavy-tailed** distribution so elites occasionally land in weak pools but usually sort high — closer to rare "superstar on weak program" events.

Assortativity knob: width of the assignment kernel (or attachment strength) controls how **assortative** mixing is — still need many pools vs few bins.

Empirical basketball: pools are **observed** `(team_id, season)` after real sorting (recruiting, transfers, minutes). §4–§6 **measure** LOO Q and peer SD **within** that realization; they **do not** re-simulate assignment.

Thread B — mean × SD (empirical diagnostic)

- **x:** LOO mean peer performance → `congestion_quality` (same object as primary Q / `ALEX_L_COL` when `PRIMARY_POOL_MODE='quality'`).
- **y:** LOO sample SD of teammate `perf` → `peer_perf_sd_loo`.
- **Color / z:** mean `Y_draft` (or promotion rate in other domains).

Hypothesis (talking point): low peer SD → **crowding** → stronger inverted-U; high peer SD → easier **distinction** → weaker crowding read. **Ranking alone drops SD information** — motivates B alongside mean-only Q.

538 note: 4B/4C fix **x** to `congestion_quality` even when `PRIMARY_POOL_MODE='crowding'` (crowding toggle affects §6 ladder cells, not this PD11 plot).

Thread C — promotion rule (simulation)

Old composite (conceptual): $w \cdot \text{local\_rank} + (1-w) \cdot A_i$.

PD11 push: weight should reflect **how much better than the pool mean**, not "#1 in pool" regardless of spread — e.g. terms involving $A_i - \bar{A}_{\text{pool}}$ (discussion: $w(A_i - \bar{A}_{\text{pool}}) + (1-w)A_i$ and algebraically related forms). Map to `537` score / weight construction before investing in **A**.

Separate exploratory idea (later): promotion may depend on **own ability × pool context** (high/mid/low A strata do not add up to the aggregate U-shape) — not the same as **A**, but related to interaction terms in the outcome model, not pool construction per se.

Where each thread lives (do not merge notebooks)

Layer	Pools	PD11 threads
`538`	**Empirical:** fixed panel (`team_id`, `season`). **Generative (planned):** soft assignment calibrated to 530	**B** (4B/4C) + §6 ladder; **A** when generative cells land
`537`	**Frozen** legacy sim — sort-and-chop + Cell 10 widgets	**C** (promotion score); compare old assignment only
`530` **CELL 5–9**	Real panel forensics	Mean/SD histograms, mean×SD, interval overlap
`535` / `tier_1_roadmap.md`	Pipeline + heavy EDA on same realized pools	Mechanism columns; not generative reassignment

Config files: `sports/sim_config.py` = **537 only**. `sports/tier1_sim_config.py` = generative defaults for **538**. Design note: `sports/documents/Tier1_Presorting_Design_Note.md`.

`537` **anchors (legacy, do not extend for Thread A):** `assign_pool_ids`, `sorting_signal_for_pools`, `SORTING_NOISE_SD`; Cell 10 (`cell10_playground_run.py`, `cell10_knob_catalog.py`, `sim_config.py`).

How §6A relates to §7 and §9

- **§6 /** `538`**:** minimal **reduced form** on **realized** pools (Wang replay) — still required for Alex conversations and cross-domain comparability.
- **§6A /** `538` **(generative):** new **assignment rules** calibrated via `530`; `537` unchanged for comparison.
- **§9:** cross-domain program keeps the **same §6 ladder** per domain; **§6A** is the shared **assignment + promotion rulebook** to align Army / academia / sports simulations before claiming universality.

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7. Where simulation fits (`537` vs `538`)

- `538` **(empirical):** §4–§6 ladder on **real** `(team_id, season)` pools — primary Alex deliverable.
- `538` **(generative, planned):** soft assignment + replay **530-style** overlap checks; knobs in `tier1_sim_config.py`.
- `537` **(frozen):** legacy sort-and-chop lab + Cell 10 widgets; use only to compare **old** assignment or to try **promotion-score** (Thread C) tweaks via `sim_config.py`.
- `530` **CELL 5–9:** forensic targets for calibrating generative assignment (not simulation).

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8. Deliverables for the next conversation

- This outline filled in at **talking-point** depth (sub-bullets, not dissertation length).
- Matching cells in `538_alex_tier1_model_and_fit.ipynb` so figures/tables follow sections 4–6.

9. Wang-style program — same skeleton, many domains

Alex's sequence is how you **lock a minimal empirical model** in one domain. The Wang-line aspiration in your notes is: **(i)** a stable empirical shape, **(ii)** a **small** generative story, **(iii)** extra **testable** predictions, **(iv)** replay in other domains.

Keep constant across domains (the "minimal model that does the trick"):

Layer	What stays the same	What changes per domain
Unit	Individual (or role-holder) embedded in a **local pool** at time t; outcome is **globally** scarce.	Definition of pool (team, unit, department) and calendar.
Objects	Y, amalgamated L (peer/rivalry proxy), Λ_t, baseline A.	How you **measure** L, Y, Λ from logs.
Estimand	Turning point L^* (or equivalent in z-score then mapped back).	Binning / windowing rules; institutional noise.
Fitting ladder	Bins → quadratic in L (LPM) → logit/probit → one decomposition at a time.	FE structure (season vs fiscal year vs cohort).

How you *find* minimality (not wish it):

1. **Start with the briefing's Tier 1 spec** — one L proxy, $L + L^2 + A$, report L^*. That *is* the candidate minimal reduced form.
2. **Add complexity only on forensic rules:** e.g. crowding or minutes enters only if (a) theory says the shape should move with that margin *and* (b) the data move that way, or the minimal spec is badly misspecified in a pre-registered sense. Your "one diagnostic at a time" rule is the operational version of minimality.
3. **Use `537` for Wang-style *comparative statics*** without bloating the empirical notebook: e.g. shift effective Λ, concentration, or opportunity; check whether L^* / top-bin behavior moves in the direction `Vector_Questions` lists (shift of peak, stronger top decline under congestion). That ties "simple mechanism" to **extra predictions**, not only the inverted-U picture.
4. **Per new domain:** copy the **same section headings** (this doc §1–§6) into a short domain addendum or notebook; replace §4's table with that domain's column mapping; rerun the ladder. The **abstract** story should read almost unchanged; only the measurement bridge changes.

Honest scope: "All domains" is **parallel replay of the same checklist**, not one pooled mega-regression on day one. The Wang papers earn universality by showing the **same qualitative mechanism** with domain-specific codings; your Alex outline is the contract for what must match across those codings.

10. `538` implementation map (keep in sync with notebook)

Notebook: `sports/538_alex_tier1_model_and_fit.ipynb`
Switchboard: CELL 0 (`RUN_CELL*`, `PERF_METRIC`, `PRIMARY_POOL_MODE`, `COMPUTE_WEIGHTED_CROWDING`, …).

Outline §	Step / object	`538` cells	Notes
§4	Data map	**0 → 1 → 2 → 3**	Same pipeline as `535` cells 1–3; no heavy 535 CELL 4 EDA.
§6.1	Descriptive bins	**4, 4A**	**4** = table; **4A** = binned draft rate vs mean L in bin.
§6A (PD11-B)	Q × peer SD diagnostic	**4B, 4C**	Associational; see **§6A**. Gated by `RUN_CELL4B`.
§6.2	Quadratic LPM + L^*	**5**	Overlay: binned means + fitted LPM ($\tilde{A} = 0$).
§6.3	Logit + L^*	**6**	Same z-scores as **5**; needs `statsmodels`.
§6.4	Inference / robustness	**7+**	Not implemented yet.
§6A (PD11-A)	Generative pool assignment (planned)	`538` + `tier1_sim_config.py`	Calibrate to **530** CELL 8–9.
§6A (PD11-C)	Promotion score experiments	`537` Cell 10, `sim_config.py`	Legacy only; optional compare to 538.
§7	Forensics / legacy sim	**530** CELL 5–9; `537` frozen	Empirical ladder in **538** §4–§6.

When you add or rename a `538` **cell:** update this table and the **§6** bullets above.

What this document is *not*

- **Not "PD11 = only the heatmap."** PD11's **center of gravity** is **pre-sorting / pool construction** (**§6A thread A**, planned in `538`; `537` **frozen**). The **Q × peer SD** block (**4B / 4C**, **§6A thread B**) is one empirical check; **promotion vs rank** (**§6A thread C**) can still be probed in **537** Cell 10.
- **Not "538 solves pool formation." 538** §6 assumes **realized** `(team_id, season)` pools; see **§6A**.
- **Not the same as** `tier_1_roadmap.md`. That file is the **living checklist for** `535` (pipeline contract, `df` columns, 535 CELL 0). **538** follows *this* outline §4–§6; the roadmap *links* here but still centers **535** for mechanism EDA and exports.

- **Spine for Alex meetings:** §1–§6 + §10 (**538** empirical). **Generative rulebook: §6A** + `Tier1_Presorting_Design_Note.md` + `tier1_sim_config.py` (**538**, when coded). **Legacy sim: 537** only.

Changelog

Date	Note
2026-05-18	`Tier1_Presorting_Design_Note.md`; `tier1_sim_config.py`; generative → **538**, **537** frozen; `sim_config` 537-only.
2026-05-18	**§6A:** PD11 pool formation, promotion rules, 538 vs 537 split; §3/§7/§10/"What this is not" cross-refs.
2026-05-18	§10: `538` cell map; §4/§6 cross-refs; clarify vs `tier_1_roadmap` and Paper Directions 11.
2026-05-12	§9: Wang-style cross-domain program — same skeleton, measurement bridge per domain; `537` for comparative statics.
2026-05-12	Initial skeleton: Alex-sequenced spine; links to existing briefing + new `538` notebook.